RS minimal model: floor summary and literature reproductions
working notes, June 2026. Quark sector, minimal (non-custodial) Randall-Sundrum.

Quick notes after the constraint-code audit and fixes. Two things here: (1) the floor summary (what $\mathrm{M_{KK}}$ the minimal model actually needs), and (2) our reproductions of the relevant published figures, side by side with the paper. Everything is the minimal (non-custodial) RS model in the S1 anarchic scenario, so in each comparison the paper panel on the left is cropped to the SINGLE minimal/S1 panel we actually reproduce (the custodial panels and the other scenarios in those figures are deliberately not shown). All run on the fixed code.

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1. Floor summary

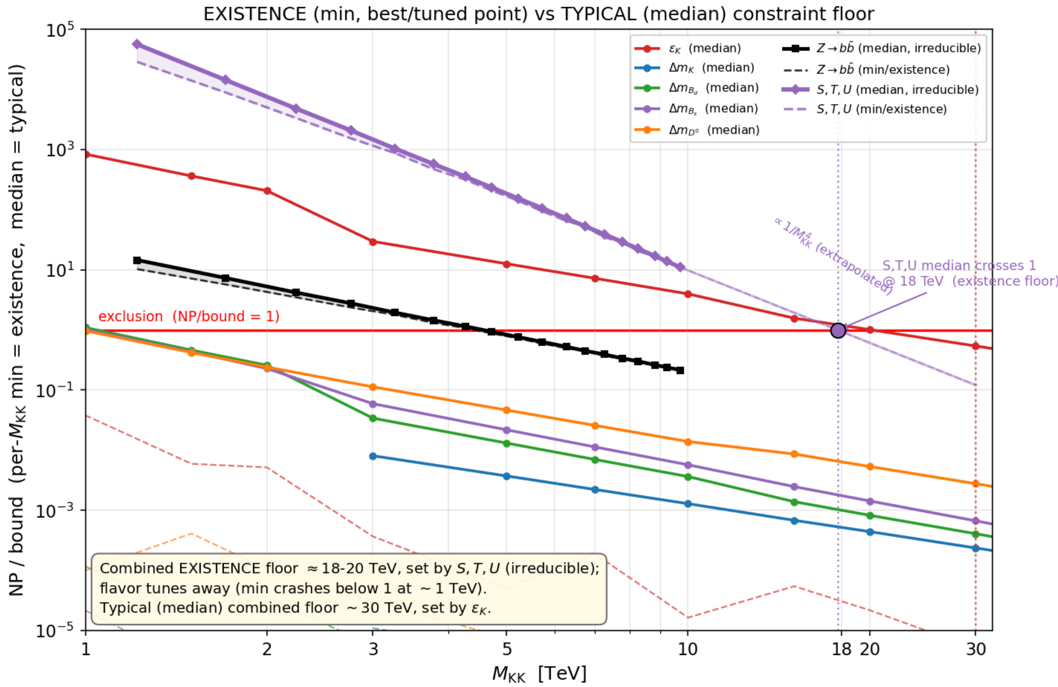
There are two honest notions of a floor, and they are very different:

- **Typical floor:** where the *median* anarchic point becomes allowed. This is the usual literature statement. With current data it is about **30 TeV**, set by ε_K.
- **Existence floor:** where even a single best-case (fine-tuned) point is allowed. This is about **18 to 20 TeV**, set by S, T, U.

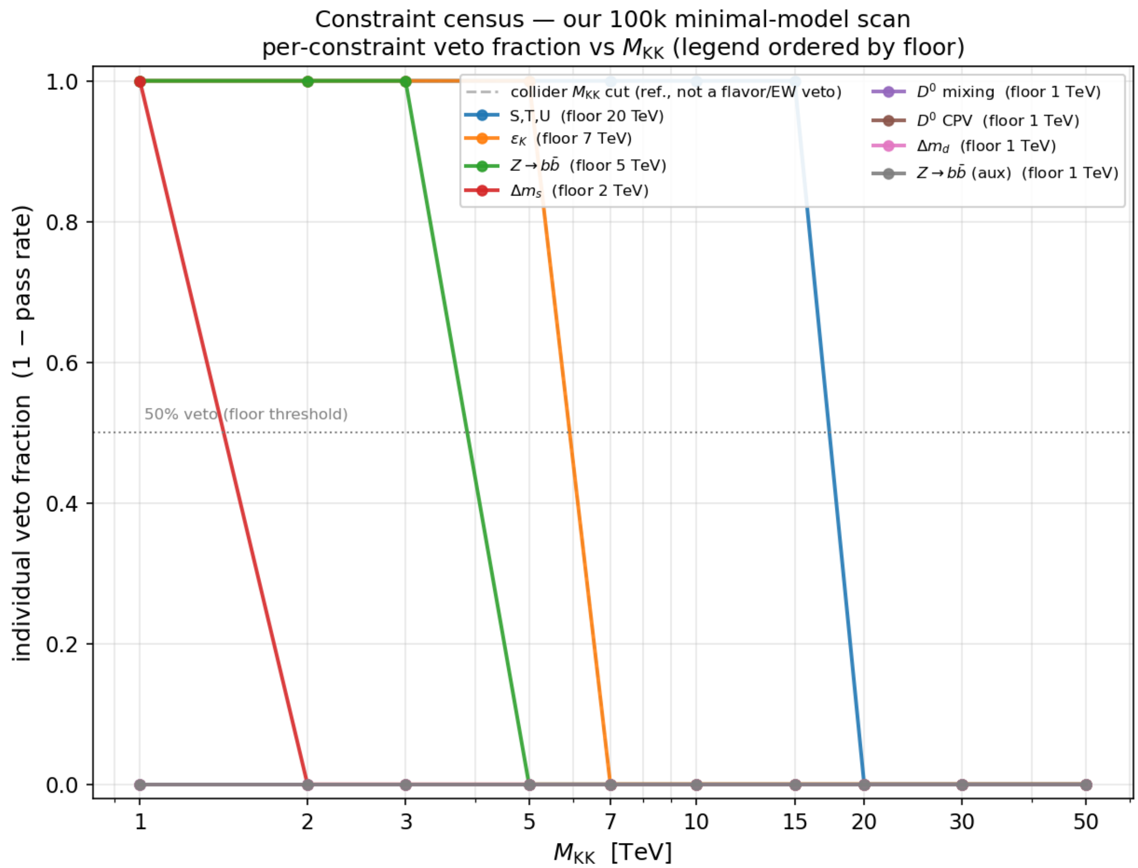
The split matters because the flavor constraints are tunable (you can align phases so the new-physics piece cancels, e.g. $\varepsilon_K \propto \operatorname{Im} M_{12} \to 0$), but S, T, U is not: in minimal RS the oblique T depends only on M_{KK} and the warp volume, with no Yukawa freedom. So even a fine-tuned minimal RS point cannot sit below about 18 to 20 TeV, and that wall is the oblique S, T, U bound. This is exactly why custodial RS exists (it protects T).

Constraint	Typical (median) floor	Existence (best/tuned) floor	Tunable?
ε_K	$\sim$30 TeV (current)	below $\sim$1 TeV	yes (align $\operatorname{Im} M_{12} \to 0$)
Δm_d (B_d)	few TeV	below $\sim$1 TeV	yes
Δm_s (B_s)	few TeV	below $\sim$1 TeV	yes
D^0	low	below $\sim$1 TeV	yes
Δm_K	low	below $\sim$3 TeV	yes
$Z \to b\bar{b}$ (T010)	$\sim$5 TeV	$\sim$4.6 TeV	no (gauge, c_{Q_3} set by m_t)
S, T, U (EW001)	$\sim$20 TeV	$\sim$18 to 20 TeV	no (no Yukawa dependence)
Combined	**$\sim$30 TeV** (ε_K)	**$\sim$18 to 20 TeV** (S, T, U)	

Existence vs typical, per constraint. Min ratio-to-bound (best, fine-tuned point) and median ratio (typical point) vs M_{KK}. The flavor curves crash well below 1 (tunable, no real floor); S, T, U has min $\approx$ median and crosses 1 only near 18 to 20 TeV (irreducible).



Our 100k scan, constraint census. Per-constraint veto fraction vs M_{KK} from our minimal scan with all fixes in. Ranking: S, T, U at 20, ε_K at 7, $Z \to b\bar{b}$ at 5 (this was the old 25 to 30 TeV artifact before the fix), Δm_s at 2. Dropping $Z \to b\bar{b}$ does not move the combined floor, so it is not the driver.



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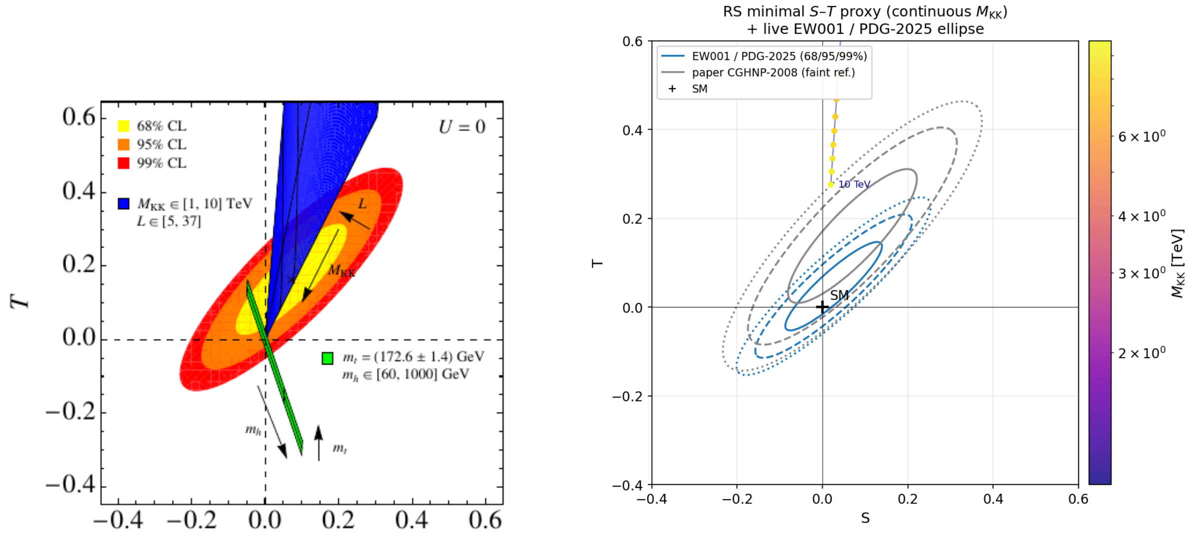
2. Reproductions (paper left, ours right)

Each comparison is the correctly cropped single paper panel (minimal RS / S1 anarchic) on the left, our reproduction on the right. Where the published figure was a multi-panel (a custodial companion, a Higgs-mass sweep, or the S2/S3/S4 scenarios), only the one panel we reproduce is shown.

S, T, U oblique (Casagrande et al. 0807.4937, Fig. 4)

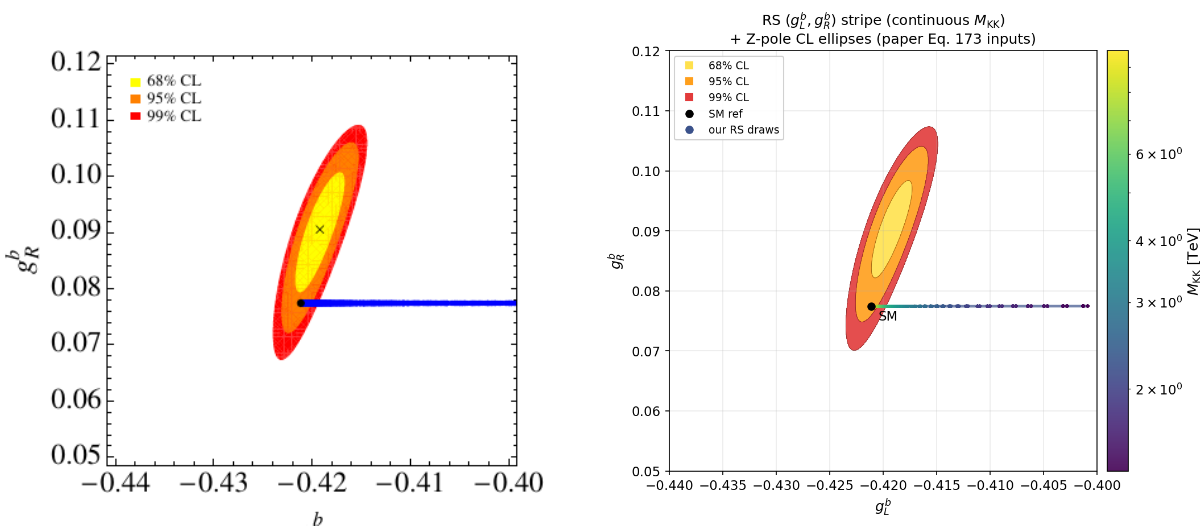
Our S-T plane vs the paper. The RS minimal wedge climbs steeply in T (the volume-enhanced T problem) and leaves our PDG-2025 ellipse fast; grey is the 2008 ellipse for reference. This is the irreducible bound that sets the existence floor. The paper panel on the left is the minimal RS one (the blue wedge shooting up in T); the custodial companion panel, where the blue band stays pinned near $T = 0$, is not what we model and is dropped.



paper (left, 0807.4937 Fig. 4, minimal RS panel) vs ours (right)

$Z \to b\bar{b}$ couplings (Casagrande et al. 0807.4937, Fig. 8)

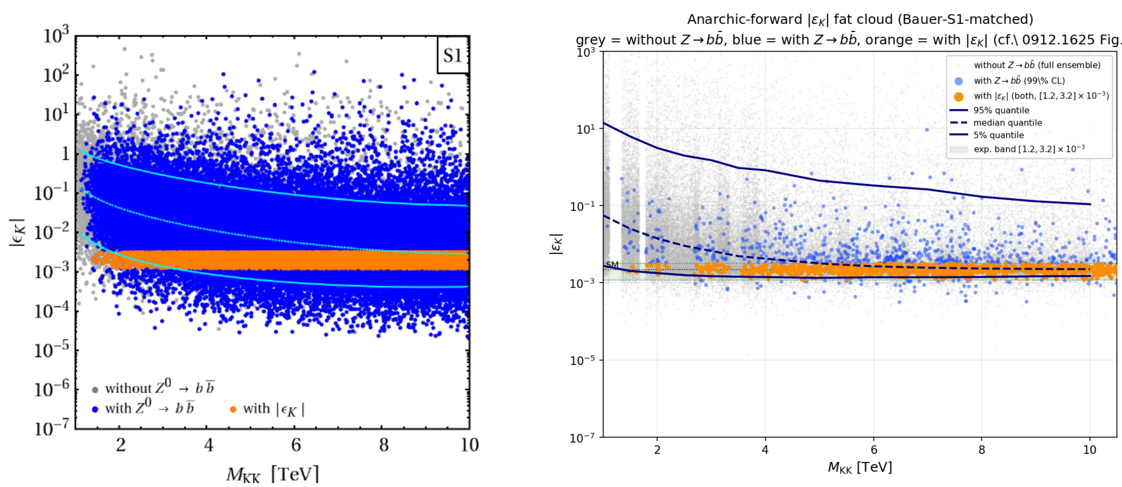
Our (g_L^b, g_R^b) plane vs the paper. The RS stripe pins at $g_R^b \approx 0.077$ and slides to less-negative g_L^b as $M_{\rm KK}$ drops, staying right by the SM point. With the B1 fix this is gauge-dominated and weak (bites only to about 5 TeV), not the 25 to 30 TeV the bug produced. The paper panel on the left is the RS scatter/stripe; the companion $m_h = 60$ to 1000 GeV Higgs-mass sweep panel, which we do not compute, is dropped.



paper (left, 0807.4937 Fig. 8, minimal RS panel) vs ours (right)

ε_K (Bauer et al. 0912.1625, Fig. 4)

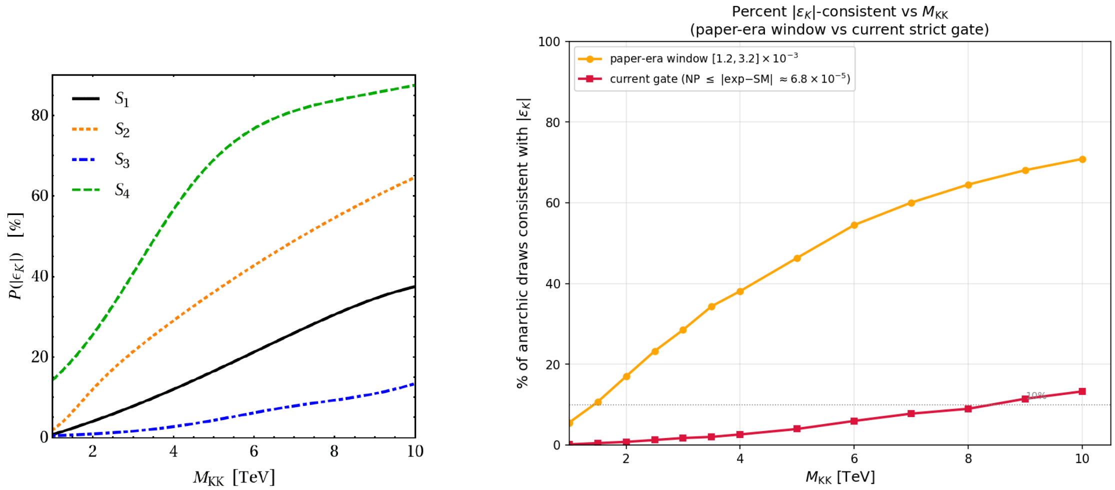
Our anarchic $|\varepsilon_K|$ cloud vs the paper, using Bauer's S1 draw setup ($Y_{\max} = 3$, scanned c). Fat, about 6 to 7 decades, with the 5/50/95 percent quantile curves. The 95 percent quantile enters the allowed band near 10 TeV, which is Bauer's headline number. Following Bauer's Fig. 4 caption the points are coloured by $Z \to b\bar{b}$ consistency: grey is the full ensemble without the $Z \to b\bar{b}$ cut, blue is consistent with both $Z \to b\bar{b}$ couplings at 99% CL, and orange is consistent with both $Z \to b\bar{b}$ (99% CL) and $|\varepsilon_K| \in [1.2, 3.2] \times 10^{-3}$ (95% CL). The per-draw $Z \to b\bar{b}$ shift is the total mass-basis $[2, 2]$ entry (gauge-KK plus the Casagrande fermion-KK admixture) computed on the same anarchic draws by `scripts/anarchic_bauer_s1_zbb.py`. The paper figure is a 2x2 over scenarios S1 to S4; the left panel here is the S1 anarchic panel (top-left of the original), the only one we reproduce, and S2/S3/S4 are dropped.



paper (left, 0912.1625 Fig. 4, S1 panel) vs ours (right)

Consistency fraction (Bauer et al. 0912.1625, Fig. 5)

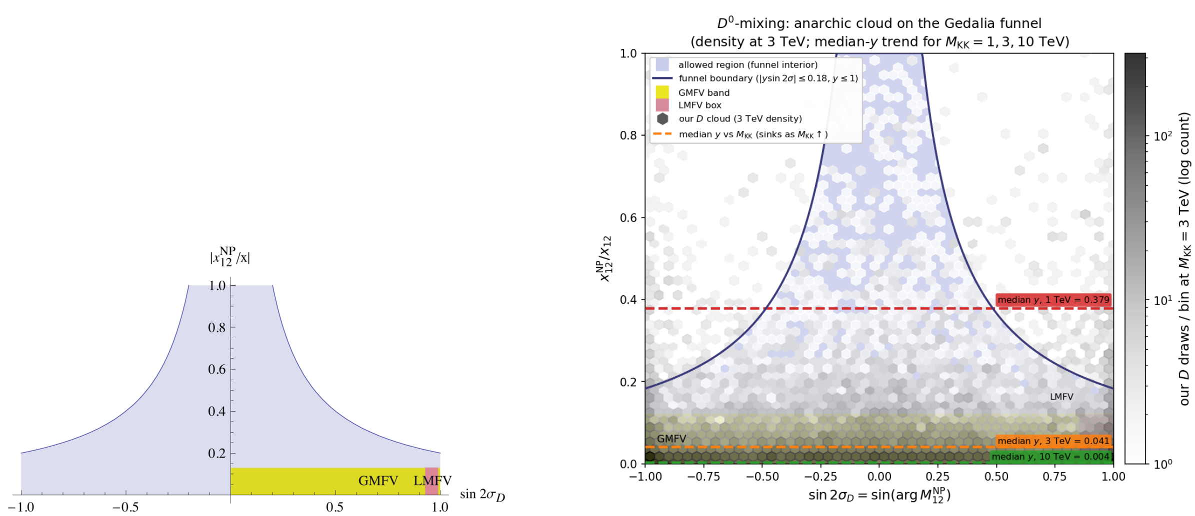
Fraction of anarchic points consistent with $|\varepsilon_K|$ vs M_{KK}, paper-era window vs current strict gate. The rising $1/M_{\text{KK}}^2$ decoupling is the same shape as the paper. The paper figure has three panels, $P(|\varepsilon_K|)$, $P(Z \to b\bar{b})$, and $P(\text{total})$; the left panel here is the $P(|\varepsilon_K|)$ one (top-left of the original), since that is the quantity we plot. We reproduce the S1 (black solid) curve in it; the $Z \to b\bar{b}$ and total panels are dropped.



paper (left, 0912.1625 Fig. 5, $P(|\varepsilon_K|)$ panel) vs ours (right)

D^0 mixing (Gedalia, Grossman, Nir, Perez 0906.1879, Fig. 1)

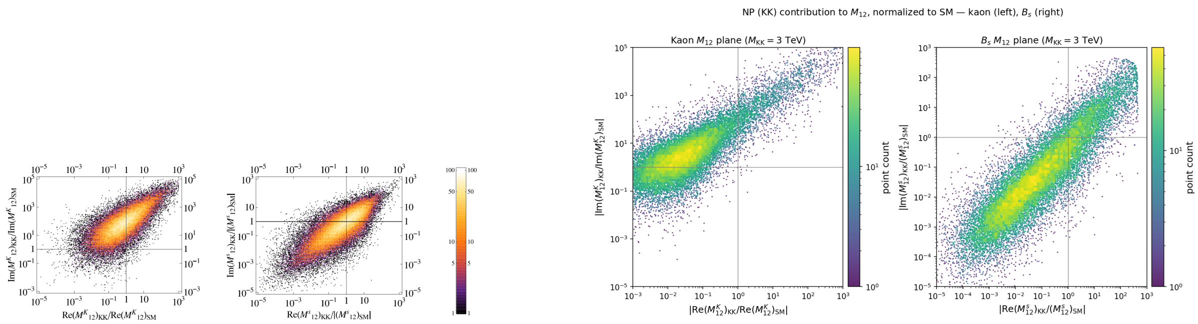
Our anarchic D-meson cloud on the Gedalia funnel vs the paper, cleaned up to a single 3 TeV density plus the median-y trend at 1, 3, 10 TeV. The cloud sinks into the allowed funnel as M_{KK} rises. The paper figure is a single panel (the allowed funnel with the GMFV and LMFV bands), kept as is. (Optional figure: say the word if you would rather drop it.)



paper (left, 0906.1879 Fig. 1, single funnel panel) vs ours (right)

$\Delta F = 2$ amplitudes (Blanke et al. 0809.1073, Fig. 2)

Re vs Im of the KK contribution to M_{12}, for K (left sub-panel) and B_s (right sub-panel). For kaons the imaginary part dominates the real part by about $100\times$ (the ε_K problem); ours is $106\times$. The paper Fig. 2 is exactly these two systems, K and B_s (it has no D or B_d sub-panels), so both are kept on the left.



paper (left, 0809.1073 Fig. 2, K and B_s panels) vs ours (right)

Dropped on purpose: the B_d and B_s phase planes (the old Figs 6 and 7). Our model clouds are fine, but the experimental CL contours we drew there are only Gaussian approximations of the CKMfitter regions, not the exact published contours, so they are not a clean reproduction. Left out rather than shown as if exact.